

Engineering Mechanics

Mumbai University Engineering Syllabus - Pre-loaded Reference for Conscious Commute.

Module 1: System of Coplanar Forces

Classification of force systems, principle of transmissibility, composition and resolution of forces, resultant of coplanar force systems.

Module 2: Equilibrium of Coplanar Forces

Conditions of equilibrium, free body diagrams, Lami's theorem, equilibrium of beams.

Module 3: Friction

Laws of friction, static and dynamic friction, wedge friction, ladder friction, belt friction.

Module 4: Kinematics of Particles

Rectilinear motion, motion curves (a-t, v-t, s-t), curvilinear motion, projectile motion.

Module 5: Kinetics of Particles

Newton's second law of motion, D'Alembert's principle, equation of dynamic equilibrium.

Module 6: Work, Power, Energy & Impulse-Momentum

Work energy principle, power, potential and kinetic energy, law of conservation of energy. Impulse-momentum principle, conservation of momentum, collision of elastic bodies.